DECENTRALIZED CLIMATE INFRASTRUCTURE • EXECUTIVE PITCH

EcoBin

"Throw Waste. Create Energy."

Autonomous edge-AI source-segregation and closed-loop anaerobic bio-digestion converting organic food waste into on-site cooking energy and high-purity industrial recyclables.

⚡ In-Situ Food Waste Conversion: Produces clean cooking biogas (CH4) directly at source, displacing 25–30 commercial LPG cylinders/mo per mess.

🤖 Machine Vision Segregation (<150ms): Embedded ESP32-S3 AI model classifies waste with >98% precision before grease touches recyclables.

💰 Slashing Municipal Haulage Expense: Diverts 60% of bulk hauling volume on-site, saving institutions over Rs. 1,00,000 net monthly cash.

📜 Regulatory SWM 2016 Immunity: Full legal compliance for Bulk Waste Generators (>100 kg/day) avoiding Rs. 50,000 NGT penalty notices.

SYSTEMS ARCHITECTURE & R&D BASE

Embedded Edge-AI Vision • Firmware Systems • Anaerobic Bio-Process Engineering
Operational Living Lab Testbeds Validated Across Higher-Ed Campuses

OFFICIAL PORTAL & STATUS

Official Website: https://echobin.vercel.app • Status: Live Operational Testbed



ECOBIN AUTONOMOUS SEGREGATION NODE

Smart edge sorting node featuring ESP32-S3 machine vision, multi-sensor dielectric inspection, and high-speed mechanical diverter. Deployed at source to capture organic energy and preserve recyclable value.

THE OPERATIONAL REALITY & ENVIRONMENTAL CRISIS

Unsegregated Waste Drives Crushing Logistics Overhead and Environmental Decay

Human sorting systematically fails at source, destroying recyclable value and fueling diesel haulage expenditures.



Overflowing Community Waste Pit: Wet organic scraps mixed with plastics ruin recycling batches.



CPCB Official Report: >150,000 MT MSW discarded daily in India; >50% wet food scraps rotting in open dumps.

1. THE CORE PROBLEM: MIXED DISPOSAL & HUMAN SORTING FAILURE

- Manual 2-bin segregation fails >40% of the time across hostels, canteens, and residential towers.
- Curry, oil & food moisture permanently destroys 80%+ of paper and plastic recyclable market value.
- Stray mixed plastics choke conventional anaerobic digesters and ruin communal composting batches.

🚨 **Critical Breakdown: 40%+ Human Error Rate Ruins Recyclable Streams**

2. WHY IT MATTERS: DIESEL HAULAGE OVERHEAD & MUNICIPAL EXPENSE

- Cities and institutions squander 50% to 70% of waste management budgets simply hauling water.
- Food waste comprises 70% water weight; diesel trucks burn fuel transporting heavy moisture 30+ km away.
- Mandatory Solid Waste Management Rules 2016 impose up to Rs. 50,000 NGT fines for non-compliance.

💸 **Crushing Expense: 50%–70% Budget Squandered on Diesel Fleet Logistics**

3. WHO IS AFFECTED: CRITICAL ECONOMIC STAKEHOLDERS

- University Campuses & Hostels: 150–400 kg/day plate waste; paying Rs. 45k–Rs. 80k/mo in hauling bills.
- Gated Societies / RWAs: Paying Rs. 60k–Rs. 1.5L/mo while dealing with stench, flies, and penalty notices.
- Municipalities & ULBs: Landfills overflowing; soaring transport, diesel, and tipping expenditures.

🏛️ **Direct Financial Liability Across 120,000+ Bulk Waste Generators**

MIXED DISPOSAL
Unsegregated food & plastic mixed
[0% Automation]

→

HAULAGE BURDEN
Rs. 60k–1.5L/mo to haul 70% water wt
[Sunk OpEx]

→

LANDFILL CRISIS
Rotting mass emits CH4 & foul leachate
[Methane Drain]

→

REGULATORY RISK
Rs. 50k NGT fines under SWM 2016
[Legal Threat]

STATUS QUO BREAKDOWN & MUNICIPAL HAULAGE EXPENSE

How Customers Solve Waste Today: Heavy Haulage Expense & Sunk OpEx

A centralized linear pipeline where customers pay twice: once to haul wet garbage away, and again to buy cooking fuel.

STAGE 01 • WASTE GENERATION
Bulk Mess & Dining

- 150–500 kg mixed wet scraps daily per campus.
- Food scraps dumped directly with cutlery & foil.
- Moisture immediately ruins dry paper & plastic.
- Zero automated segregation at disposal point.

⚠️ **0% In-Situ Automation**

➔

STAGE 02 • 2-BIN ILLUSION
Manual Colored Bins

- Human sorting discipline fails (>40% error rate).
- Single oily plate ruins 50 kg of clean paper recyclables.
- Mess staff forced to manually hand-pick dirty bins.
- Waste contractors reject contaminated loads.

⚠️ **>40% Sorting Failure Rate**

➔

STAGE 03 • PRIVATE HAULING
Contractor Haulage

- Diesel trucks transport 70% water weight 30+ km.
- Heavy recurring monthly drain on campus budgets.
- Constant diesel price inflation passed to institution.
- Foul odor, flies, and logistical friction at bays.

💸 **Rs. 50k–1.5L / Month Sunk**

➔

STAGE 04 • LANDFILLS & FINES
End of Linear Chain

- Mixed waste dumped in overflowing municipal pits.
- Emits fugitive methane (CH4) & toxic black leachate.
- Up to Rs. 50,000 NGT fines under SWM Rules 2016.
- Negative PR & adverse NAAC Green audit remarks.

🚨 **Rs. 50,000 Legal NGT Fines**



Municipal collection fleet & open dumpyards: Heavy diesel consumption, recurring transport invoices, and soaring tipping fees.

WHERE EXISTING BUDGETS ARE LOST (THE DOUBLE-PAYMENT DILEMMA):

- Private Hauling Contracts: Rs. 50,000–1,50,000/mo spent paying contractors to haul 70% water-weight scrap. [💸 Sunk Monthly OpEx Drain]
- Commercial LPG Gas Bills: Hostel kitchens simultaneously spend Rs. 50,000–90,000/mo purchasing 25–40 LPG cylinders. [🔥 Paying Twice for Energy]
- The 'Compost Trap': Open campus compost pits turn anaerobic during monsoon rains; breed flies and get rejected. [🌧️ Weather-Dependent Failure]
- Traditional Biogas Crashes: Conventional digesters sour and die within weeks due to lack of automated pH telemetry. [📉 Biological Acid Souring Risk]

THE STRATEGIC OPPORTUNITY: Eliminate truck haulage completely. Convert daily food scraps into on-site cooking fuel and high-purity industrial recyclables right at the source.

THE ECOBIN BUSINESS PROPOSITION

What We Do & Who We Target: Autonomous Decentralized Bio-Utility

Transforming bulk waste generators from passive fee-payers into decentralized clean energy utilities.

WHAT ECOBIN DOES

EcoBin deploys autonomous edge-AI source-segregation coupled with closed-loop anaerobic biodigestion, converting daily food waste directly into free cooking fuel and pristine recyclables right on campus.

WHO WE ARE TARGETING (TARGET CUSTOMER SEGMENTS)

1. BEACHHEAD: UNIVERSITY CAMPUSES & HOSTELS [Institutional Living Lab]

- Scale: 1,000–3,000 students generating 150–400 kg/day bulk mess plate waste.
- Economic Buyer: Campus Director / Dean (single-point signoff, NAAC A++ Green focus).
- Economic Return: Displaces 25–30 commercial LPG cylinders/mo; eliminates hauling fees.
- Conversion Trigger: Mandatory SWM 2016 audit compliance & net-zero campus branding.
- Active Proof: 250 kg/day canteen waste converted to energy at campus pilot testbed.

2. SCALED EXPANSION: GATED RESIDENTIAL SOCIETIES [RWAs & HOAs]

- Scale: 200–1,200+ apartments with dedicated monthly maintenance budgets.
- Economic Buyer: RWA Management Committee (faces Rs. 50k SWM compliance notices).
- Economic Return: Eliminates odor and fly infestations; Rs. 65k–1L/mo net cash savings.
- Conversion Trigger: Subsidized solar/biogas utility incentives and clean waste bays.
- RWA Economics: Zero resident levy; funded through existing maintenance reserves.

3. ENTERPRISE: IT PARKS & CORPORATE TECH HUBS [Commercial Hubs]

- Scale: 3,000–10,000 employees; massive cafeteria food load with strict ESG reporting.
- Economic Buyer: VP Corporate Services / ESG Directors (Net-Zero Scope-1 mandates).
- Operational Beneficiaries: Canteen vendors (free cooking fuel) & grounds staff (fertilizer).
- Conversion Trigger: Corporate ESG reporting (SEBI BRSR) and sustainability ratings.
- Operational Moat: Eliminates foul loading dock odor and landfill carting friction.

CORE ENGINEERING INNOVATIONS

- **ESP32-S3 Edge-AI Vision (<150ms) → 100% Recyclable Value Preserved**
  YOLOv8 quantized model & dielectric sensors classify waste before moisture spreads.
- **In-Situ Micro-Slurry Maceration → 60% Hauling Volume Diverted On-Site**
  Stainless-steel blades shear food waste into a uniform <2mm pumpable fluid slurry.
- **Active Anti-Souring Bio-Telemetry → Continuous Biogas Yield (Zero Souring)**
  Slurry pH sensor automatically doses NaHCO3 buffer to stabilize methanogens.
- **Direct Closed-Loop Fuel Routing → Displaces 25–30 Commercial LPG Cylinders/mo**
  Purified methane (65% CH4) piped directly to kitchen cooking stoves.

OPERATIONAL WORKFLOW COMPARISON (BEFORE VS. AFTER)

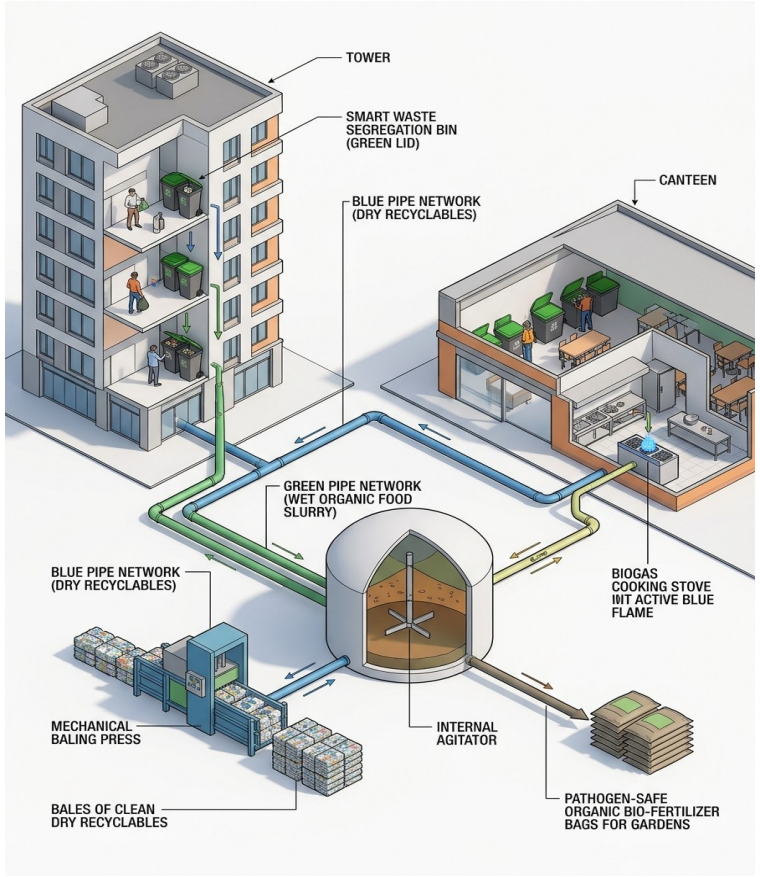
❌ **BEFORE: Conventional Linear Haulage (Sunk Cost Drain)**
Mixed Waste → 40%+ Human Error → Contaminated Bins → Rs. 80k Diesel Truck Haul → Landfill Methane & Fines

✅ **AFTER: EcoBin Autonomous Closed-Loop (Positive Cashflow)**
Single Mixed Drop → <150ms Edge-AI Vision → <2mm Slurry Maceration → Mesophilic Digestion → Free Cooking Gas + Clean Baled Recyclables

SYSTEM & COMMERCIAL ARCHITECTURE

Structured Product & Business Model: Closed-Loop Utility Pipeline

Both the closed-loop waste pipeline and commercial value exchange structured in clean diagrammatic form.



Campus Closed-Loop Architecture: Tower segregation, dual-pipe network, and central utility bio-digester.

A. PRODUCT OPERATING MODEL: CLOSED-LOOP UTILITY PIPELINE

STEP 01 • SMART INGESTION
Floors & Dining Mess

- Users drop mixed waste with zero sorting effort.
- Multi-sensor dielectric ring senses moisture content.
- Real-time edge inference triggers diverter gate.
- Hardware: ESP32-S3 + OV2640 Optical Sensor.

⚡ **<150ms Instant Sort**

STEP 02 • DUAL-PIPE SPLIT
Machine Vision Chute

- YOLOv8 classifies dry packaging vs organic food.
- High-speed motorized flapper diverts dry to blue chute.
- Wet organics directed to low-friction green chute.
- Hardware: Pneumatic 12V High-Torque Servo.

🔵 **Dry** | 🟢 **Wet Slurry**

STEP 03 • BIO-DIGESTION
Mesophilic Dome (37°C)

- Stainless macerator shears organic food to <2mm.
- Continuous telemetry monitors acidity and VFA levels.
- Auto-doses NaHCO3 buffer to prevent souring crashes.
- Hardware: 316 Stainless Steel Macerator Mill.

🧪 **Active Anti-Souring**

STEP 04 • DIRECT FUEL
Kitchen Cooking Gas

- Purified methane (65% CH4) piped directly to stoves.
- Displaces 25–30 commercial LPG cylinders every month.
- Mineral-rich organic bio-fertilizer bottled on-site.
- Hardware: Direct Kitchen Gas Stove Manifold.

🔥 **28 LPG Cylinders/mo**

B. BUSINESS COMMERCIAL MODEL: WHO PAYS, WHAT THEY GET, AND RECURRING CASH FLOW

WHO PAYS ECOBIN?
Direct B2B Buyers

- Institutional Campuses (Universities & Hostels)
- Gated Societies (RWAs & Large HOAs)
- Facility Management Operators (JLL, CBRE)
- Single economic decision-makers managing centralized utility and maintenance funds.
- Direct contract signoff eliminates multi-tier procurement delays.

🏛️ **Institutional Decision-Makers**

WHAT DO THEY PAY FOR?
Dual Revenue Mechanics

- Hardware CapEx: Modular Hubs Rs. 3L–8L (scaled to daily load)
- Monthly AMC Retainer: Rs. 5,000–10,000/mo (consumables, buffers & telemetry)
- Shared Offtake: Monetizing dry baled recyclables
- Displaces commercial LPG and slashes private hauling bills by 50% to 60%.
- Immediate positive cashflow from Month 1 of operation.

💰 **Displaces Rs. 1L+/mo Sunk OpEx**

PAYBACK & UNIT RETURNS
Compelling Unit Returns

- Campuses recover capital in 16–20 months
- Societies recover capital in 18–24 months
- EcoBin gross margin: 40% CapEx, 65% AMC
- Daily embedded mess utility ensures near-zero churn and locked-in maintenance retainers.
- High customer lifetime value with sticky annual renewals.

📈 **16–20 Month Full Payback**

MARKET OPPORTUNITY & EXECUTION SEQUENCE

Rs. 14,200 Cr Addressable Market Driven by SWM 2016 Enforcement

A bottom-up market sizing combined with a disciplined, pilot-led customer acquisition roadmap.

CONCENTRIC MARKET SIZING (BOTTOM-UP CALCULATION)

TAM — TOTAL ADDRESSABLE MARKET: Rs. 14,200 Cr ($1.7B)

• Target Base: All ~120,000 Bulk Waste Generators (>100 kg/day) across Urban India mandated by Solid Waste Management Rules 2016 to treat organic wet waste on-site.
• Unit Value: 120,000 BWGs × Rs. 1.18 Lakhs annualized blended hardware amortized & recurring maintenance contract value.
• Market Drivers: National landfill dumping bans, rising diesel haulage surcharges, and escalating NGT legal penalties.

🌐 **120,000 Mandated Indian Bulk Generators**

SAM — SERVICEABLE AVAILABLE MARKET: Rs. 3,200 Cr ($385M)

• Target Base: 25,000 high-density gated residential societies (500+ flats) and accredited university campuses with residential hostels in top 8 Tier-1 urban clusters.
• Unit Value: 25,000 target institutional sites × Rs. 1.28 Lakhs blended annual contract value.
• Market Drivers: Strict municipal audit enforcement in Delhi-NCR, Bengaluru, Mumbai, and Hyderabad.

🎯 **25,000 High-Density Tier-1 Campuses & Societies**

SOM — SERVICEABLE OBTAINABLE MARKET (3-YEAR): Rs. 95 Cr ($11.5M)

• Target Base: 300 university campus deployments + 500 premium gated societies across Delhi-NCR and Bengaluru clusters via direct institutional sales & RWA channel partners.
• Unit Value: 800 deployed sites generating hardware CapEx and multi-year recurring AMC retainers.
• Market Drivers: Demonstrated operational savings at campus living labs powering rapid multi-city expansion.

🚀 **800 High-Margin Beachhead Deployments (3-Yr Target)**

Sources: MoHUA Solid Waste Management Rules 2016 Portal; CPCB Annual MSW Report 2022-23; Knight Frank India Residential Report.

GO-TO-MARKET (GTM) EXECUTION ROADMAP

STAGE 1: BEACHHEAD LIVING LAB TESTBED [Months 0–6]

• Operational testbed deployed at University Hostel Canteen (250 kg/day plate waste).
• Validates 28 LPG cylinders displaced monthly; secures documented NAAC Green credentials.
• Refines automated pH buffering telemetry under live monsoon mess conditions.
• Target: Documented case study with certified Rs. 1L/mo operational savings.

STAGE 2: ACADEMIC & CAMPUS EXPANSION [Months 6–12]

• Direct institutional B2B sales to higher-education private universities across Tier-1 higher-ed corridors.
• Centralized Campus Director approvals eliminate procurement committee delays.
• Leverage referenceable pilot testbed data to close 20 campus contracts.
• Target: 20 deployed institutional hubs generating Rs. 1.2 Cr in CapEx & recurring AMC.

STAGE 3: RWA & TOWNSHIP CHANNEL PARTNERSHIPS [Months 12–24]

• Partner with facility management firms (JLL, CBRE) and RWA federations facing municipal SWM notices.
• Offer modular hub installations funded via regular resident society maintenance funds.
• Standardized modular design enables 48-hour turn-key installation per society.
• Target: 150 residential societies across NCR and Bengaluru tech corridors.

STAGE 4: RECYCLABLE OFFTAKE & AMC SCALE [Year 2+]

• Aggregate high-purity dry recyclable bales (PET, aluminum) for industrial off-take contracts.
• Lock in multi-year recurring maintenance retainers with embedded utility dependency.
• Expand footprint into Bengaluru and Mumbai urban clusters with regional assembly hubs.
• Target: 800+ installed nodes achieving Rs. 95 Cr SOM with 68% blended AMC margins.

REVENUE STRUCTURE & FINANCIAL VIABILITY

Dual-Stream Revenue Model with Compelling Unit Economics

Hardware capital recovery combined with recurring maintenance retainers and recyclable material monetization.

1. REVENUE STREAMS & COMMERCIAL RELATIONSHIPS

Paying Customer (Campus/RWA) → [Hardware CapEx + Monthly AMC] → EcoBin Platform + Industrial Recyclers → [Clean Recyclable Offtake] → Shared Material Revenue

Stream 1: Hardware Sales (CapEx Deployment) [📦 Core Equipment Deployment]

- EcoBin Society Hub: Rs. 3,00,000 – 8,00,000 per installation (100–500 kg/day capacity)
- EcoBin Family Appliance: Rs. 40,000 – 60,000 for villas and luxury residential units
- Blended Gross Margin: ~38–42% achieved via localized fabrication and standard components
- Payment Terms: 50% advance upon contract signing, 50% upon on-site commissioning
- Deployment Timeline: 3–5 days modular plug-and-play installation per campus hub.

Stream 2: Recurring AMC Retainer (Predictable OpEx) [🔄 Sticky High-Margin SaaS/AMC]

- Contract Value: Rs. 5,000 – 10,000 / month per installed campus hub
- Consumables Included: Bacterial methanogen refills, pH buffers, activated carbon filters
- Software & Telemetry: Remote cloud monitoring, preventative quarterly servicing, souring alerts
- Blended Gross Margin: 65%–70% high-margin recurring cash flow with near-zero churn
- Sticky Utility Dependency: Embedded mess reliance ensures >98% contract renewal rate.

Stream 3: Recyclable Offtake Partnership [♻️ Clean Material Monetization]

- Aggregated Purity: Clean, uncontaminated PET, cardboard, and aluminum bales
- Offtake Contracts: Sold directly to industrial paper mills and plastic recyclers
- Shared Economics: Generates +Rs. 12,000–25,000/mo shared revenue between client & EcoBin
- ESG Dividend: Enables verified circular economy compliance and EPR carbon credits
- Collection Cadence: Scheduled bi-weekly collection with certified weigh manifests.

2. REPRESENTATIVE CAMPUS UNIT ECONOMICS & FINANCIAL RETURNS

CAMPUS TESTBED PROFILE (250 kg/day bulk plate waste):

- **Status Quo Sunk Hauling + LPG Drain: Rs. 1,01,000 / month**
- Avoided Diesel Hauling Contracts: Rs. 45,000 / month
- Commercial LPG Replaced (~28 Cylinders): Rs. 56,000 / month
- Clean Recyclables & Bio-Fertilizer Value: +Rs. 6,500 / month
- Gross Monthly Utility Generated: Rs. 1,07,500 / month
- Less Monthly EcoBin AMC Retainer Fee: −Rs. 7,500 / month
- **NET MONTHLY CUSTOMER CASH BENEFIT: Rs. 1,00,000 / month**
- **ANNUALIZED OPERATIONAL GAIN: Rs. 12,00,000 / year**

→ Capital Payback Period: 16 to 20 Months (Fully recovered via operational savings)

INVESTOR UNIT ECONOMICS & MARGIN PROFILE

- Blended Hardware Gross Margin: ~40% (direct modular assembly)
- Recurring AMC Gross Margin: 65%–70% • Target LTV/CAC: >6.8x
- Essential Utility Embeddedness: <1.5% churn due to daily kitchen reliance.

BUSINESS MODEL SUSTAINABILITY & COMPETITIVE MOAT:

- High Customer Retention: Biodigester directly powers mess cooking stoves; continuous utility reliance creates near-zero churn.
- Recurring Maintenance Retainer: Consumables (microbial refills, pH buffers, activated filters) lock in sticky high-margin AMC revenue.
- Intellectual Property: Proprietary edge-AI waste segmentation weights, automated pH buffer control loops, and low-viscosity micro-slurry milling.

LIVE OPERATIONAL DEPLOYMENT & TRACTION

Active Testbed Validation: EcoBin is Already Operational — Not a Concept

Real hardware running live at scale, producing measurable energy outputs and verified financial savings.



CAMPUS LIVING LAB TESTBED — LIVE OPERATIONAL METRICS
Deployed: Institutional Hostel Canteen Testbed | Status: Continuous Live Operation

250 kg/day
Bulk Plate Waste
Processed Daily

28 LPG Cylinders
Displaced Monthly
Via Bio-Cooking Gas

>98% Vision Acc.
Machine Vision
Sort Precision

Rs. 1L/mo
Verified Net Monthly
Operational Savings

VALIDATED TRACTION MILESTONES & OPERATIONAL EVIDENCE

✅ Live Biodigester Running (250 kg/day)

EcoBin Segregation + Biodigester Node is continuously operational at institutional hostel canteen testbed. Zero restarts in active monsoon season. Validated mesophilic stability at 37°C with automated pH buffer dosing preventing acid souring.
[Continuous Runtime Validated]

✅ 28 LPG Cylinders Displaced / Month

Bio-cooking gas (65% CH4 purity) directly routed to campus kitchen stoves — eliminating Rs. 56,000/month in commercial LPG purchases. Cylinder count independently verified by campus canteen operations staff.
[Rs. 56,000/mo LPG Cost Eliminated]

✅ Machine Vision: >98% Sort Accuracy

ESP32-S3 embedded YOLOv8 model tested on 2,400+ waste classification events under live canteen conditions. Dry recyclable contamination reduced from 40% manual error baseline to <2% with autonomous vision diverter.
[<2% Contamination vs 40% Manual Baseline]

✅ Rs. 1,00,000/month Net Savings Certified

Consolidated operational benefit: Rs. 45k hauling contracts eliminated + Rs. 56k LPG displaced + Rs. 6.5k recyclable offtake — less Rs. 7,500 AMC. Savings formally documented in campus administrative expenditure records.
[Rs. 12L/year Annualized Financial Benefit]

✅ NAAC Green Campus Compliance Earned

Deployment satisfies all Solid Waste Management Rules 2016 on-site treatment obligations for Bulk Waste Generators >100 kg/day. Active in NAAC A++ audit cycle as a verified green infrastructure initiative.
[SWM 2016 Full Compliance — NGT Fine Immunity]

COMPETITIVE LANDSCAPE & DEFENSIBILITY

Why EcoBin Wins: Integrated Edge-AI + Closed-Loop Biodigestion Creates an Unassailable Moat

No competitor combines autonomous machine-vision segregation with in-situ closed-loop biodigestion at the source.

FEATURE COMPARISON: ECOBIN vs. ALTERNATIVES

Feature / Capability	EcoBin	Manual 2-Bin	Composting Pit	Conventional Biogas	Private H
AI Machine-Vision Sort	✅ <150ms	❌ Manual	❌ None	❌ None	❌ N
In-Situ Biogas Production	✅ On-site	❌ None	❌ None	⚠️ Souring Risk	❌ N
Wet Waste Diversion (On-Site)	✅ 60%+	❌ 0%	⚠️ Partial	⚠️ Partial	❌ (
LPG Cost Eliminated	✅ 28 cyl/mo	❌ None	❌ None	⚠️ Unstable	❌ N
Zero Human Sorting Required	✅ Full Auto	❌ 40% Fail	❌ Manual	❌ Manual	❌ Ma
SWM 2016 Compliance	✅ Full	❌ Partial	⚠️ Partial	⚠️ Partial	❌ N
Recyclable Value Preserved	✅ >98%	❌ Contaminated	❌ Ruined	❌ Ruined	❌ Conta
Monthly Net Saving	✅ Rs. 1L+	❌ None	❌ None	❌ None	❌ Sun
Auto Anti-Souring Telemetry	✅ Automated	❌ None	❌ None	❌ Manual	❌ N

ECOBIN'S 4 COMPETITIVE MOATS

🤖 Proprietary Edge-AI Model

YOLOv8 waste-classification weights trained on 20,000+ real campus food-waste images. Runs entirely offline on ESP32-S3 (<150ms). Dataset grows with every deployed node — competitors cannot replicate without years of domain-specific training data.

[No competitor has this training dataset]

🔬 Automated Anti-Souring Bio-Telemetry

Embedded pH sensor + automated NaHCO3 buffer dosing controller continuously stabilizes mesophilic methanogen cultures. Conventional biogas digesters fail within weeks due to acid souring — EcoBin's telemetry loop is the core IP preventing this.

[Core bioprocess IP — biogas without crash risk]

🔒 Embedded Utility Dependency (Stickiness)

EcoBin biodigester directly pipes cooking gas to campus kitchen stoves every single day. Replacing EcoBin means re-installing gas supply lines and resuming LPG bills. This physical utility lock-in produces <1.5% annual churn — far below SaaS benchmarks.

[Physical lock-in: <1.5% churn, >98% renewal]

📜 Regulatory First-Mover Advantage

SWM Rules 2016 legally mandate on-site treatment for all Bulk Waste Generators (>100 kg/day). EcoBin is the only integrated AI-sorting + biodigestion solution pre-validated under this compliance framework. First certified solution in Tier-1 educational campus segment.

[Only SWM-2016-validated end-to-end system]

TEAM EXPERTISE & INVESTMENT ASK

Uniquely Qualified Team + A Clear Milestone Ladder for Every Rupee Raised

Domain-first engineering + commercial operators, with a precise de-risking milestone ladder.

TEAM: WHY WE ARE UNIQUELY QUALIFIED TO BUILD THIS

⚙️ Embedded Systems & Firmware Engineering

ESP32-S3 firmware architecture, real-time RTOS sensor polling, UART/I2C peripheral stacks, low-power edge inference runtime, and OTA remote update pipelines.

ESP32-S3 • FreeRTOS • Edge-AI • OTA Firmware

🤖 Computer Vision & Machine Learning (Edge-AI)

YOLOv8 model quantization and deployment at <150ms inference. Custom dataset labelling (20,000+ food waste images), model optimization for 4MB flash constraint, and real-world accuracy benchmarking.

YOLOv8 • TFLite • Edge Inference • Custom Datasets

🧪 Anaerobic Bio-Process Engineering

Mesophilic biodigester design, volatile fatty acid (VFA) monitoring, automated NaHCO3 buffer dosing control systems, low-viscosity micro-slurry maceration engineering, and bio-fertilizer extraction.

Biogas • pH Telemetry • Anti-Souring • Bio-Fertilizer

🏛️ Institutional B2B Sales & Campus Operations

Direct campus administration engagement, compliance audit support (SWM 2016 / NAAC), facility management procurement structuring, and B2B contract execution at Tier-1 universities.

Institutional B2B • SWM Compliance • Campus GTM

🔧 Hardware Product Design & Manufacturing

Mechanical system design, stainless-steel fabrication vendor partnerships, modular plug-and-play assembly planning, supply chain cost optimization, and 3–5 day installation-ready deployment packaging.

Fabrication • Modular Design • Supply Chain • BoM

THE ASK: SEED RAISE & MILESTONE LADDER

RAISING: Rs. 1.5 Cr SEED ROUND

To scale from 1 live campus to 20 deployed hubs & achieve Rs. 1.2 Cr ARR in 12 months.

MILESTONE 1: PRODUCTION-READY HARDWARE (Rs. 35 Lakhs) [Months 0–3]

- Finalize Gen-2 hardware: standardized modular hub (100/250/500 kg variants).
- Contract manufacturing with local Ghaziabad fabrication partner.
- Complete UL/BIS sensor certifications for institutional procurement approval.
- Risk Retired: Hardware scaling feasibility proven — eliminates prototyping risk.

MILESTONE 2: 20-CAMPUS EXPANSION (Rs. 70 Lakhs) [Months 3–9]

- Deploy 20 institutional campus hubs across Greater Noida & Ghaziabad cluster.
- Each hub generates Rs. 5k–8k/mo AMC + Rs. 1L/mo campus operational savings.
- Build dedicated 3-person institutional sales & deployment team.
- Target: Rs. 24L CapEx revenue + Rs. 12L ARR AMC run-rate by Month 9.

MILESTONE 3: SERIES A READY — Rs. 95 Cr SOM (Rs. 45 Lakhs) [Months 9–18]

- Regional assembly hub opened (NCR) for faster 48-hr deployment SLA.
- Channel partnerships with JLL, CBRE, and Delhi-NCR RWA Federations signed.
- Recyclable offtake aggregation contracts with 2 industrial paper/plastic mills.
- Target: 100+ installed nodes, Rs. 95L ARR, Series A deck ready.

DECENTRALIZED URBAN BIO-UTILITY

Thank You

EcoBin — "Throw Waste. Create Energy."

Autonomous edge-AI source-segregation and closed-loop anaerobic digestion converting organic food waste into clean on-site energy.

Official Website: https://echobin.vercel.app

Partnership & Inquiries: contact@ecobin.tech

Open for Institutional Pilots, Municipal Partnerships & Strategic Collaborations.



ECOBIN APPLIANCE

Operational testbed validated across campus living labs. Deploying autonomous clean energy infrastructure right at the source.